

CHANDIGARH UNIVERSITY

University Institute of Computing (UIC)

**PROJECT REPORT ON
E-Commerce Web-Application Using MERNSTACK**

Submitted by

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MASTER OF COMPUTER APPLICATION

Under the Guidance of Mrs.

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OBJECTIVES

1. To develop a full-stack E-Commerce platform

- The main objective is to build a complete web application using the MERN stack (MongoDB, Express, React, Node.js).
 - It should handle both frontend and backend operations efficiently.
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2. To implement user authentication and authorization

- To provide secure login and registration using JWT (JSON Web Token).
 - To ensure only authorized users can access certain features.
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3. To create a role-based system (Seller & Buyer)

- To differentiate users into **Seller** and **Buyer** roles.
 - Sellers can add, update, and delete products.
 - Buyers can view products and add them to the cart.
-

4. To design a user-friendly interface

- To create a clean, responsive, and easy-to-use UI.
 - To improve user experience through modern design and navigation.
-

5. To manage products dynamically

- To allow sellers to manage product listings.
- To store product details such as name, price, category, and images in the database.

INTRODUCTION

In today's digital era, e-commerce has become an important part of everyday life. People prefer online shopping because it is convenient, time-saving, and provides a wide variety of products. Businesses are also shifting towards online platforms to expand their reach and improve customer experience.

This project, “**MERN Market – E-Commerce Website**”, is developed to create a simple and efficient online shopping platform using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. These technologies help in building a complete full-stack web application.

The system allows users to register as either a **buyer** or a **seller**. Sellers can manage products by adding, updating, and deleting them, while buyers can browse products, view details, and add items to their cart. The application also ensures secure login using JWT authentication and follows role-based access control.

The project includes important features such as product listing, search and filtering, image upload, and a user-friendly interface. These features make the application similar to real-world e-commerce platforms.

Overall, this project helps in understanding how frontend, backend, and database work together to build a complete web application and provides practical knowledge of full-stack development.

SOFTWARES USED

The development of this project requires various software tools and technologies to build, test, and manage the application effectively. The following software has been used:

1. Frontend Technologies

- **React.js (Vite):** Used to build a fast and interactive user interface. It helps in creating reusable components and improves user experience.
 - **HTML & CSS:** Used for structuring and styling the web pages.
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2. Backend Technologies

- **Node.js:** Provides a runtime environment to execute JavaScript on the server side.
 - **Express.js:** Used to build RESTful APIs and handle server-side logic efficiently.
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3. Database

- **MongoDB:** A NoSQL database used to store user data, product details, and other application information in a flexible format.
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4. Authentication & Security

- **JWT (JSON Web Token):** Used for secure user authentication and maintaining login sessions.
 - **Bcrypt.js:** Used to hash and protect user passwords.
-

5. Development Tools

- **Visual Studio Code:** Code editor used for writing and managing project files.
 - **Postman:** Used for testing backend APIs.
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6. Version Control

- **Git & GitHub:** Used for version control and project management.

Input's Screenshots

×

Welcome to Shopy.

Login

Register

Email

john@example.com

Password

.....

Login

Figure 1 Login Form

×

Welcome to Shopy.

Login

Register

Name

Email

Password

I want to

Buy Products

Sell Products

Register

Figure 2 Registration Form

OUTPUT SCREENSHOT

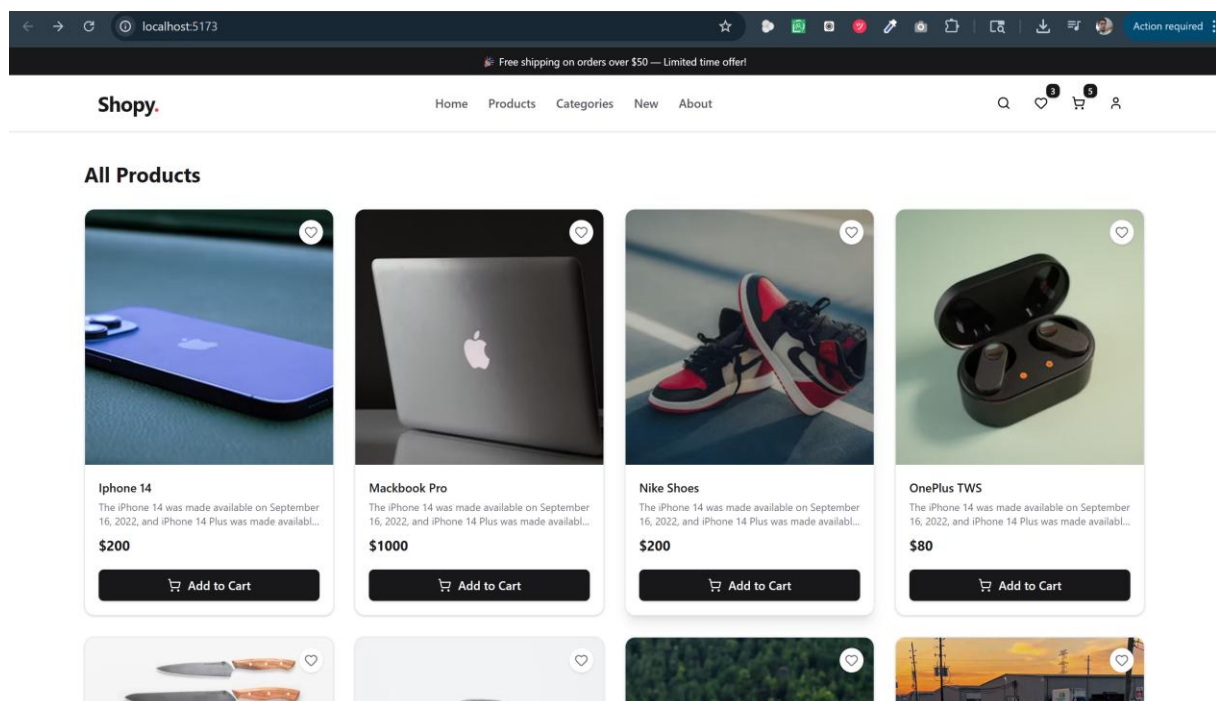


Figure 3 Website Products Page

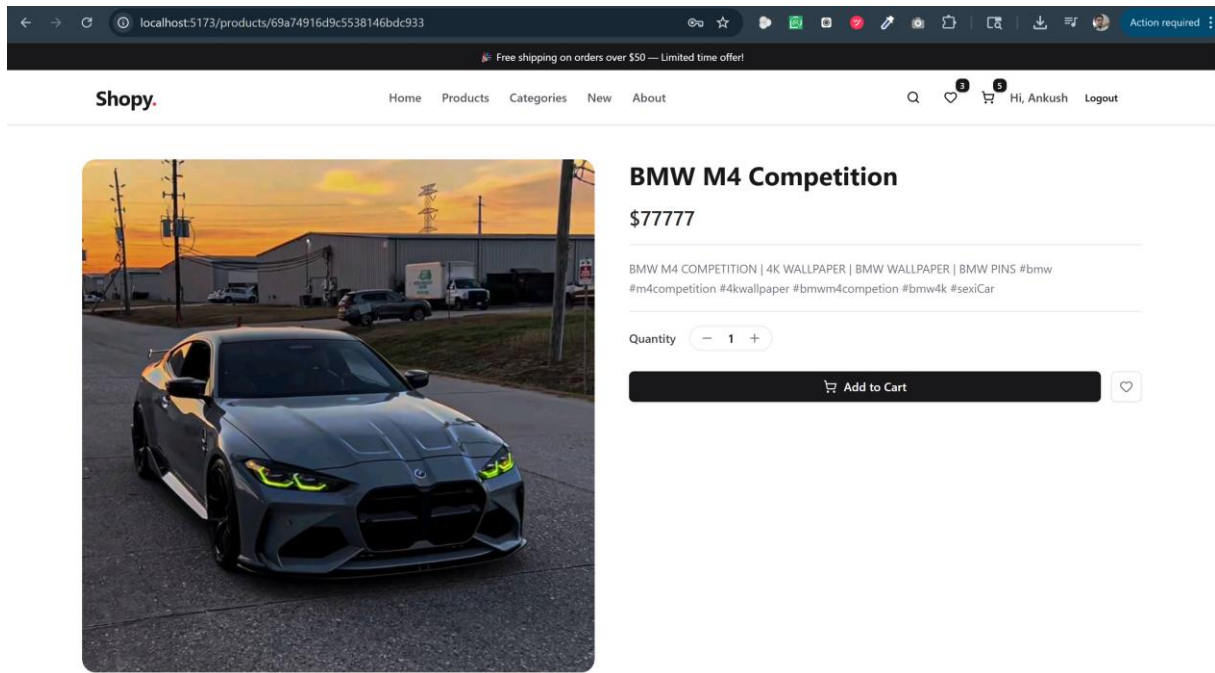


Figure 4 Product Description Page

CODE SCREENSHOTS

```
const productRoutes = require('./APIs/productRoutes');
const authRoutes = require('./APIs/authRoutes');

app.use(cors({
  origin: ['http://localhost:5173'],
  credentials: true
})))
app.use(express.urlencoded({ extended: true }));
app.use(express.json());
app.use(cookieParser());

app.use(productRoutes);
app.use('/auth', authRoutes);

app.get('/', (req, res) => {
  res.send('Working Fine !');
})

const PORT = process.env.PORT || 4444;
app.listen(PORT, () => {
  console.log('Server is up at PORT', PORT);
})
```

Figure 5 Index.js

```
const productSchema = new mongoose.Schema({
  name: {
    type: String,
    trim: true,
    required: true
  },
  img: {
    type: String,
    trim: true,
    default: '../images/product.jpg'
  },
  price: {
    type: Number,
    min: 0,
    default: 0
  },
  desc: {
    type: String,
    trim: true
  },
  createdBy: {
    type: mongoose.Schema.Types.ObjectId,
    ref: 'User' // reference to User model
  }
})

const Product = mongoose.model('Product', productSchema);
```

Figure 6 Product Schema

LEARNING OUTCOMES

Through the development of this MERN-based E-Commerce Website, several important concepts and practical skills were learned:

1. Understanding Full-Stack Development

- Learned how to build both frontend and backend of a web application.
- Understood how different parts of an application work together.

2. Working with MERN Stack

- Gained hands-on experience with MongoDB, Express.js, React.js, and Node.js.
 - Learned how to connect frontend with backend using APIs.
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3. Implementation of Authentication

- Understood how JWT (JSON Web Token) works for secure login.
 - Learned how to manage user sessions and protect routes.
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4. Role-Based Access Control

- Learned how to implement different roles (Seller & Buyer).
 - Understood how to restrict access to specific features based on user roles.
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5. Database Management

- Learned how to design schemas using MongoDB and Mongoose.
 - Performed CRUD (Create, Read, Update, Delete) operations on data.
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6. API Development

- Learned how to create RESTful APIs using Express.js.
 - Understood request-response cycle and error handling.
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7. UI/UX Design

- Improved skills in designing responsive and user-friendly interfaces using React.
 - Learned to enhance user experience with better layouts and navigation.
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8. Debugging and Problem Solving

- Developed the ability to identify and fix errors in both frontend and backend.
 - Improved logical thinking and debugging skills.
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9. Real-World Application Development

- Understood how real e-commerce platforms work.
- Gained confidence in building industry-level projects.

REFERENCES

The following resources were used during the development of this project:

1. Github Link: <https://github.com/artistic-engineer-13/Ecommerce>

2. Official Documentation

- React.js Documentation – <https://react.dev/>
 - Node.js Documentation – <https://nodejs.org/>
 - Express.js Documentation – <https://expressjs.com/>
 - MongoDB Documentation – <https://www.mongodb.com/>
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3. Learning Platforms

- GeeksforGeeks – <https://www.geeksforgeeks.org/>
 - W3Schools – <https://www.w3schools.com/>
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4. Tools & Libraries

- Postman – for testing APIs
 - Visual Studio Code – for development
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5. Other Resources

- YouTube tutorials and online articles related to MERN stack development.